Latent Spaces Beyond Synthesis: From GANs to Diffusion Models

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ABSTRACT. This paper examines the evolving nature of internal representations in generative visual models, focusing on the conceptual and technical shift from GANs and VAEs to diffusion-based architectures. Drawing on Beatrice Fazi's account of synthesis as the amalgamation of distributed representations, we propose a distinction between "synthesis in a strict sense", where a compact latent space wholly determines the generative process, and "synthesis in a broad sense," which characterizes models whose representational labor is distributed across layers. Through close readings of model architectures and a targeted experimental setup that intervenes in layerwise representations, we show how diffusion models fragment the burden of representation and thereby challenge assumptions of unified internal space. By situating these findings within media theoretical frameworks and critically engaging with metaphors such as the latent space and the Platonic Representation Hypothesis, we argue for a reorientation of how generative AI is understood: not as a direct synthesis of content, but as an emergent configuration of specialized processes.

1. Introduction

As generative archives (Meyer), predictive media (Manovich), stochastic parrots (Bender et al.), or blurry JPEGs of the Web (Chiang), generative visual models are multifaceted and profoundly influential entities in the contemporary visual sphere. These metaphors, driven by the necessity to grasp the algorithms and their products, connect technical aspects to societal and cultural frameworks. They extricate the current logic of generativism (Martino, Mauro) that stems from so-called generative AI, separating the inner predictive logic of these algorithms and their foundation in minimization and statistical methods.

Each of these metaphors carries implicit assumptions about the functioning of the models and their representations, assumptions often unexamined even by the authors themselves. Conversely, these metaphors risk being 'folk theories' (Kempton; Lakoff, Johnson), rather than scientific explanations. In contemporary models, expectations are only partially fulfilled and become metaphors of metaphors, detached from their technical groundings.

Analyzing the processes of generative visual models is essential for their comprehension and that of their generated images, which now inhabit large sections of the Web. An analysis of these models is, however, complicated by what Offert and Dhaliwal call the 'black box casuistry'. The incommensurability between the representations generated within the models and human understanding is often taken as an excuse to approximate metaphorically their functioning without technical correspondence.

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An exemplar case of such 'folk theory' is linked to the latent space. The interest in this space stems from its vast affordances, offering a spatial and indexable representation of entities (Schaerf) to which visual studies and media philosophy theories of the previous half-century can easily be applied. Accounts that assign a quasi-religious role to this space of statistical banality (Crawford) are based on the functioning of Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), where the model learns to compress information from training into a single space that constitutes the starting point for generation. In these models, there is a one-to-one relationship between space and image, and the reciprocal relationships between points in the space are maintained in the generated image so that two neighboring points yield similar images. However, even within the short and recent history of generative AI, these models have witnessed a paradigm shift from VAE and GAN models to the current hype surrounding diffusion models (Ho et al.; I. J. Goodfellow et al.; Kingma and Welling).

2. Synthesis in Representations

> “This representational reality, internal to computation, is a stable (if imperfect) whole, a togetherness of distributed representations.” (Fazi, p. 1)

In a recent article, Fazi discussed the central role of *synthesis* in the deep learning revolution. Here, synthesis is not intended in the meaning of artifice (i.e., synthetic), but purely “as amalgamation, as a composing and putting together” (Fazi, p. 3) of representations. In the essay, Fazi seems to include in the definition of synthesis a unity that is a “togetherness of distributed representations” (Fazi, p. 1), thus accepting cases where the representations do not come together in a synthetic entity within the model but only achieve unity in relation to others. In this work, we build on her argument but introduce a further distinction between ‘synthetic in a strict sense’ and ‘synthetic in a broad sense’. The former refers to the representations within the model that, taken in isolation, provide the (near) complete representation of the generated output. The latter, instead, refers to representations that retain the full information only if taken holistically and as distributed elements. We claim that this difference, which we will later illustrate through examples, is at the heart of the paradigm shift in internal representations of vision generative models from GANs and VAEs to recent diffusion and autoregressive models. We reinforce this claim experimentally.

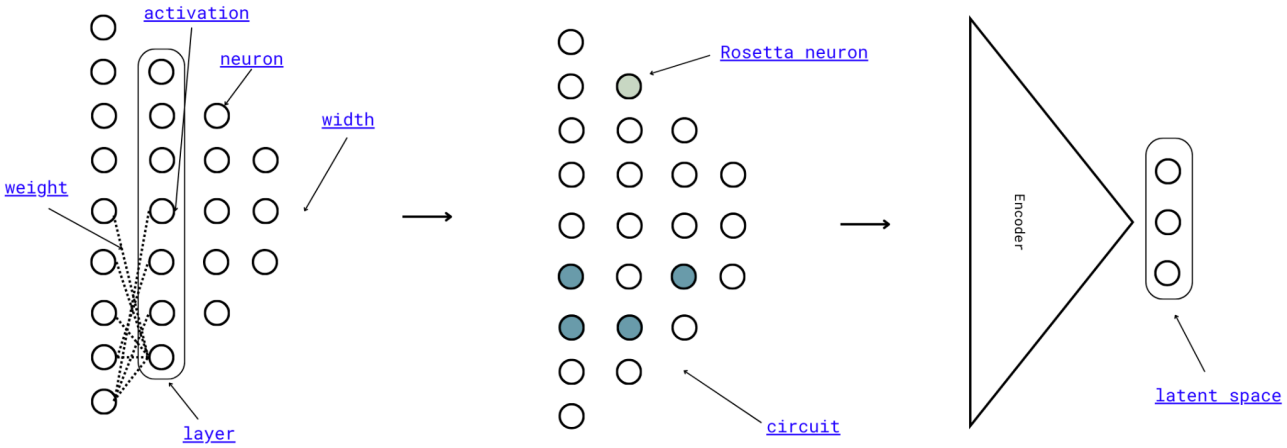
2.1. Deep Representations

To better understand this distinction, we shortly introduce the concept of representations within deep learning models. In this section, we refer to *deep representations* (or simply representations) as the internal states taken by a model while generating, in our example case, an image. This should not be confused with the *weights*, which stay the same for one model for any generation. Oppositely, the internal states, technically referred to as the *activations*, are the product of the interaction between the weights learned by the model and the particular input (in the case of diffusion models, the noise and prompt). Deep representations can be taken at different cardinalities:
- the complete set of activations can be considered one representation,

- the activations at each layer of the model constitute different representations,
- or, the activations of a specific neuron/set of neurons create a representation.

In the rest of the text, we focus on the latter two definitions, which are in practice more actionable.

In the context of deep learning, advocates for creating and understanding representations have been around since, and possibly even more so, early AI advancements (Rumelhart et al.). Bengio, in a talk in 2016, claims that creating representations within AI models is key to higher-level[2] cognition and abstraction (Bengio). Broadly speaking, deep representations are created during the model's training, often consisting of the backpropagation of errors from the prediction through the weights connecting each layer from the last to the first. The errors being backpropagated are called gradients. The learning proceeds in phases, from utility to compression. An early finding on stochastic gradient descent (SGD), a widespread training optimizer based on the gradient, shows that the learning can be interpreted from an information perspective: neural network models learn to optimize representations according to the Information Bottleneck Principle (IBP) (Shwartz-Ziv and Tishby), following which the representations are first arranged to perform best in the task and later compressed[3] to the most generalizable configuration.



Another consequence of the backpropagation is that deep representations are distributed (Offert and Impett). In fact, we do not train the single neurons to represent a specific feature of the output; on the contrary, we let the neurons collectively shape the best possible representation for the task. In practice, studies on interpretability, especially in early vision models such as Inception V3, showed that the representations of concepts can be of two kinds: single neurons or circuits (Olah et al.). In the first case, the simplest case, a neuron alone represents the entire concept, similar to the 'grandmother neuron' hypothesis from neuroscience, whereby the stimulus of seeing your grandmother would fire precisely one specific neuron. These neurons, in deep learning, are called the Rosetta neurons (Dravid et al.). A more common case is circuits of neurons, where one concept corresponds to the firing of several neurons within the model. In the former case, the representation is localized and complete. In the latter, information is distributed.

[2] By the term high-level Bengio refers to representations that are manipulated at the conscious level and verbalizable.

[3] For more details on compression in Offert and Impett.

Deep representation, intended at this cardinality of the neuron, is central to interpretability research, currently pioneered by Anthropic, which tries to find representations of concepts within models to achieve more trustworthy AI. This field of study is often tightly linked with neuroscience and the representations within the brain.

2.2: Modes of Representation

We switch now to the second cardinality of representation: the layer[4]. This cardinality is most commonly researched in the computer science field of representation learning and is frequently used in humanities and social sciences applications and theories. Layerwise representations shift the focus from looking for a specific concept, as in interpretability, to seeing which concepts are represented where and how. The affinity with humanistic approaches appears: we are no longer interested in specific and delineated concepts; instead, we wish to construct (or apply) theories around incommensurable representations to situate them in a commensurable framework.

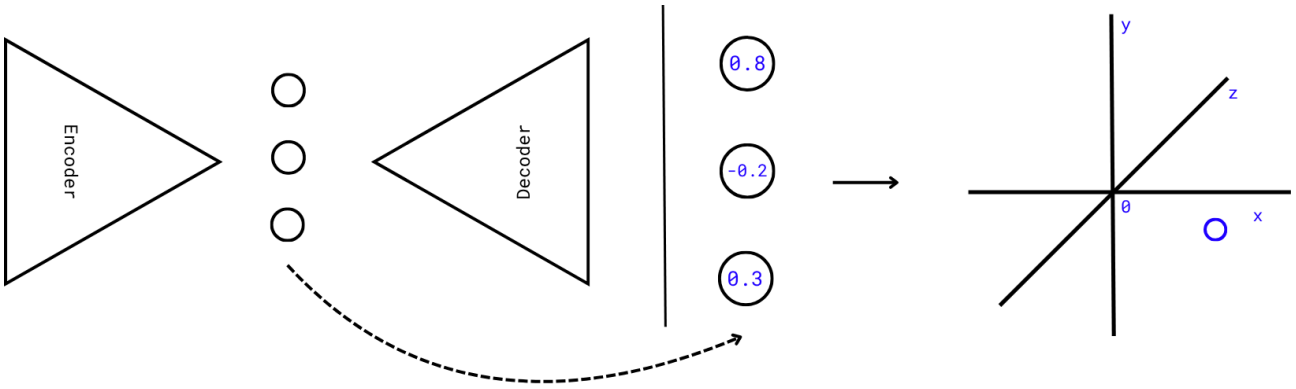
In the Deep Learning book (I. Goodfellow et al.), possibly the most widely adopted for teaching AI, the rearrangement of information within a layer, functionally to a loss, can be done in three non-exclusive ways:
- compression of information into independent representations,
- compression of information into a denser representation,
- and compression of information into a sparse representation.

Bengio, in his talk, refers to the second kind, that of disentangled or independent representations, as higher-level cognition (Schaerf, “Reflections on Disentanglement and the Latent Space”). This kind of reasoning is the closest to scientific reasoning, where the constitutive elements are pulled apart, as in neuroscience, to understand the whole. This reasoning does not consider higher-order influences, so it can be regarded as a reductionist approach (Rosenthal, David). Compression to a dense representation is the most common in modern AI and is well-supported in GPU hardware. A dense representation typically suggests that the space spanned by the neurons of that layer is continuous, and it is a manifold of the data. Lastly, sparse representations are currently in vogue for interpretability (Cunningham et al.). The simple concept behind sparse autoencoders is the possibility of untangling superposition neurons, i.e., neurons representing parts of more than one concept, into complete and informative neurons.

In the remainder of the text, we focus on dense and independent representations. After this lengthy excursus on deep representations, we are finally ready to go back to the question of synthesis.

2.3. Latent Space

[4] In this case, by layer we mean the output of a collection of neurons operating together at a specific depth within a neural network.



The most well-known example of a deep layerwise representation is that of the *latent space*. Strictly speaking, the latent space refers to the internal representation formed at the bottleneck layer of a model, where data is compressed into its most compact representation. More broadly, however, the term is often used to describe any layerwise feature representation learned by a neural network.

Consider the figure above. An autoencoder model that compresses the representation of an input image through a series of layers (encoder) into just three nodes — the smallest layer in the model, and therefore called the bottleneck. From there, the representation is expanded to reproduce the original image (decoder). The autoencoder learns by approximating the output image to be the same as the input image. The values at this bottleneck form what is called the latent space vector. Since the bottleneck contains three neurons in one row, these values define a point in a three-dimensional vector space.

The representation in this space is dense, as the bottleneck is highly compressed, and it forms a continuous space. A series of techniques and by-products[5] ensures that points in space that are close together yield images that we are inclined to consider similar[6]. Often, the representations in the latent space are also independent: not only are similar inputs yielding similar outputs, but moving the input in a specific direction in the space yields a specific variation in the output. In the example above, imagine a direction in the latent space that separates all colored images on one side and all black and white ones on the other. Moving the original point of a black and white image in that direction would yield almost the same image, but in color. These independent directions commonly occur in GANs (most commonly StyleGAN) and VAEs (bVAE).

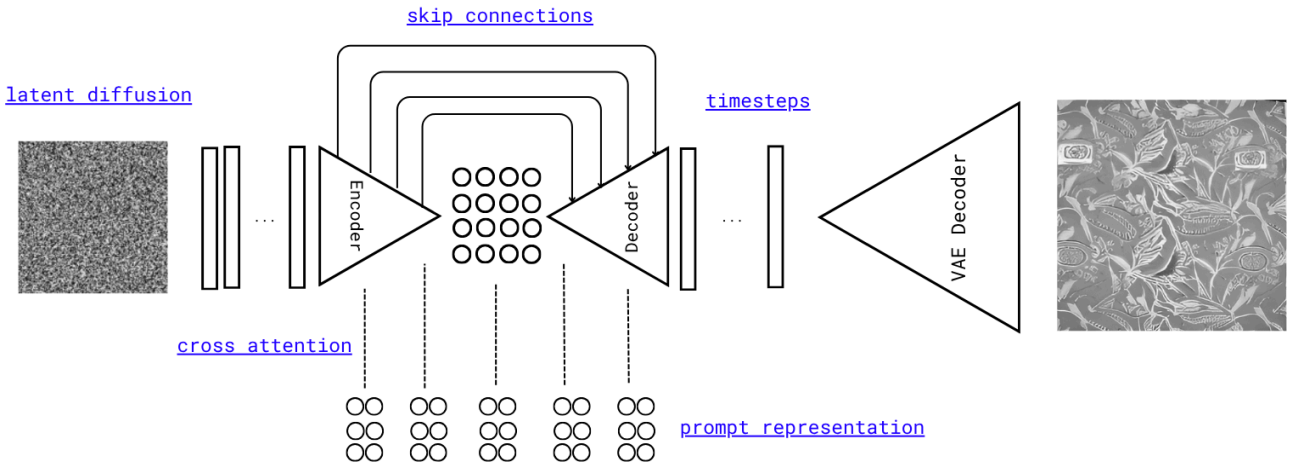
Fazi recalls Chiang's imperfect analogy of ChatGPT to a blurry JPEG of the Web. While there are clear differences underscoring the compression of representations carried out by JPEG and by transformer models (Chiang; Offert and Impett), "indirectly, the compression analogy highlights that 1) these technologies are indeed representational; 2) all representing is compressing; and 3) such compressing involves a degree of unification of information." (Fazi, p. 5) The third point is particularly indicative of our distinction: while all representing is compressing, the compression can involve different degrees of unification. In the latent space, the space spanned by the bottleneck layer of the model uniquely and entirely determines the generated output, up to some noise. The unit of re-elaboration and

[5] The chapter by Schaerf discusses the techniques that determine the continuity and semantic organization of latent spaces.
[6] Similarity is an essential concept in latent space studies, refer to Offert and Impett; Offert for more detailed discussions.

unification of the information is contained within this one space and only depends on the relationships between elements within the space. Therefore, latent spaces of these models create a ‘synthesis in a strict sense’.

The latent space as a place of synthesis has been instrumental to its widespread adoption and theoretical popularity. Nonetheless, as previously mentioned, the recent AI revolution in the visual domain was powered by a shift, around 2020, towards diffusion and autoregressive models. This shift begs the question: do these new models possess latent spaces that are synthetic in the strict sense?

3. Representations Beyond Synthesis



Diffusion models, originally introduced by Ho et al. and later popularized by Stable Diffusion and DALL·E (Rombach et al.; Ramesh et al.), have been metaphorically described as “negentropic” processes (Zylinska). Rather than reversing entropy in a thermodynamic sense, this term highlights their iterative refinement: they sculpt visual coherence from random noise over time (Salvaggio). The goal of the model is no longer to predict the image, but to predict how much noise at each timestep should be subtracted from the image, in order to carve out the content specified from the prompt. These models exhibited incredible expressivity, but their internal representations are considerably more complex than previous models.

If we consider the representations at the bottleneck layer of the model, as we did previously, we would have to consider one latent space per timestep due to the internal states changing at every iteration. Furthermore, diffusion models tend to adopt bottleneck spaces that are tensors instead of vectors: they are groups of vectors that are spatially organized. Roughly, the bottleneck could consist of 64 vectors distributed in a grid of 8x8, where the top left vector mainly contains information about the top left of the image, and so on.

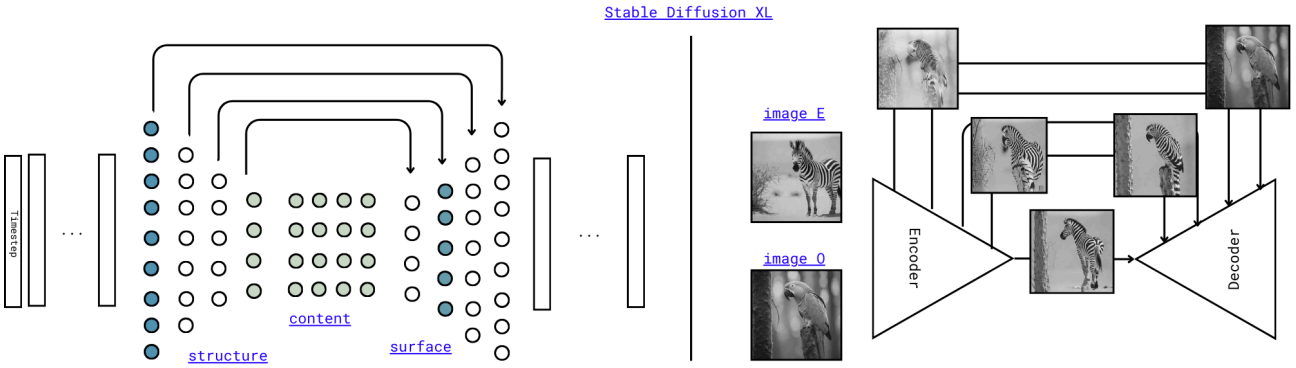
Moreover, the information in these models flows through skip connections, i.e., long-range connections between encoder and decoder. This creates a scenario, unlike previous models, where not all information passes through the bottleneck. In diffusion models, the prompt is represented through a frozen (vision-)language model (such as CLIP or T5;

Radford et al.) and incorporated into the model through cross-attention[7]. In this process, each token (roughly word/subword) is encoded separately, resulting in one representation per word.

This lack of clarity on which space could serve as the latent space shows a considerable increase in complexity in these models and prompts us to better understand these representations.

3.1: Specialization of Representation

In order to point out the layerwise representations of diffusion models, we performed the following experiment. We generated two images in parallel; we can call them image O(riginal) and image E(dit). We tested what happens when we insert the representation of image E of a specific layer into the generation of image O. We tried this using every layer in the model, carrying out the switch at every timestep. The experiments were aimed at answering two questions:
- What is the role of each layerwise representation, if there is any?
- Does the same layerwise representation apply to all diffusion models?



Through this experiment, we identified two main findings[8]. While GANs' and VAEs' latent space determined the entire image, diffusion models split the representational burden across the layers. In every model, we found one or more layers creating the *composition* of the image, one or more layers responsible for the *content*, and one or more layers responsible for *surface details and colors*. This finding was particularly interesting as it showed a tendency of more complex models with multiple information streams to create a specialization of labour.

We found these three aspects across almost all diffusion models we considered. However, the layers responsible for these aspects were found in entirely different parts of the model. Some featured the content layers after the surface layers (i.e., Stable Diffusion 1.4; Rombach et al.), while most had content layers first, and others had the most critical layers in the bottleneck and early decoder layers (i.e., Stable Diffusion XL; Podell et al.), in contrast to Stable Diffusion 1.4 and 2 that had them in the late encoder. Kandinsky even had all the important layers in the decoder (Razzhigaev et al.). This division of labour

[7] For a longer explanation of the role of the attention module in diffusion models refer to Schaerf.
[8] The complete discussion of the setup and results of these experiments can be found in Schaerf et al.

could be found even in models that do not rely on the encoder-decoder architecture, but rather on vision transformers. These last models, however, did not present layers that are specifically responsible for surface-level features.

The difference between the internal representations of these models, we believe, problematizes a popular deep learning hypothesis: the Platonic Representation Hypothesis (Huh et al.). This hypothesis argues that many large models converge to similar internal representations due to learning the structure of reality. We contest this by showing that while representational components may appear similar across models, the internal arrangement of representations remains fundamentally different.

3.2: Combination over Unity

As we have seen, the developments in vision models, most importantly the iterative process and the introduction of attention modules, removed the necessity of a unique representation, and the implications of this led to a loss of indexicality, of direct relationship to the output, and even of compression. They have created, however, a specialization of representations. The information bottleneck of previous models proved too restrictive to achieve the desired expressiveness, leaving room for even more distributed, complex, and iterative representations. We could not find any layer representation in these models that would entirely be responsible for the output, leaving behind a synthesis in a strict sense, but creating a 'synthesis in a broad sense', where the unity of representation lies in the distributed contribution of the several specialized representations in an emergent synthesis.

The Kantian definition of synthetic levels mentioned in Fazi resonates with the specialization of representations found within these models. As Fazi says: "He described 1) the synthesis of apprehension in the *intuition*, concerning elemental, raw perceptual inputs and producing temporal and *spatial structures*; 2) the synthesis of reproduction in the *imagination*, a transcendental act of the mind that unifies apprehended representations with other apprehended representations and corresponds to imagination itself; and 3) the synthesis of *recognition in the concept*, which concerns *understanding* and the application of the categories via acts of apperception. What is apprehended in intuition, reproduced in imagination, and recognized in concepts are representations (i.e., intuitions, images, and concepts are representational)." (Fazi, pp. 6-7) Kant identifies three faculties of the human mind: intuition (or perception), intellect (or episteme), and reason (or imagination) (Kant). We find that, in diffusion models, the first two faculties are reflected among the layerwise representations: some representations are mostly perceptual in nature, with the composition highlighting the spatial structuring, while others are more conceptual, referring to the contents of the images. It is the imagination, finally, that aggregates these representations into a coherent and emergent image. Here, imagination is responsible for compositionality, "a concept denoting a whole whose meaning is expressed as a function of the meanings of its combined parts" (Fazi, p.8).

The connectionist nature of neural networks, whereby meaning is created by the connections between neurons, becomes now even more structural: even the layerwise representations contribute to creating a connectionist representation.

4. Conclusion

In this essay, we proposed a conceptual and technical framework for understanding the representations of generative visual models through the lens of *synthesis*. By tracing the transition from VAEs and GANs, models that centralize meaning within a latent space, to diffusion models, which distribute representational responsibilities across layers and time steps, we redefined the nature of synthetic representation. We offered the distinction between strict and broad synthesis as a means of interpreting this shift and supported our claims with experiments into diffusion models' layerwise representations. Furthermore, we examined how attention, spatialization, and cross-modal conditioning complicate the indexical clarity of latent spaces. Our findings complicate the notion of a singular, Platonic latent representation and instead suggest a combinatorial and emergent understanding of image synthesis.

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